

Nikel: Banking on data-driven credit decisions to increase SME lending

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT NIKEL

Nikel makes fast loan decisions and new product launch half with the help of Thought Machine BigQuery.

The logo for Nikel, featuring the word "Nikel" in a bold, sans-serif font. The letter "i" is stylized with a green dot.

About Nikel

Founded in 2018, Nikel is a fintech company which enables small business lending throughout Southeast Asia. Nikel connects banks with SMEs through originators like fintech and technology providers on its digital lending platform.

Google Cloud results

- Optimizes queries for credit risk

Reduced
decision
time 1

Industries: Financial Services & Insurance

analysis at
scale with
BigQuery

week
millis

Location: Singapore

- Reduces time to market by half with each product launch, enabled by Google Kubernetes Engine and microservice-based architecture design of Vault Core
- Launched Vault Core banking platform in three months



About Thought Machine

Thought Machine powers modern finance through cloud-native banking and payments technologies. Its core banking engine, Vault Core, is trusted by leading financial institutions around the world.

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Small and medium business (SME) loans have huge potential to incite change. A farmer can increase crop yields with an effective irrigation system instead of hand watering. A furniture maker can meet production demand and grow the business by investing in woodworking equipment. In tough times, a bank loan can mean the difference between keeping workers or closing the business. However, SMEs find it hard to get funding from traditional lenders.

For banks, existing SME lending poses two challenges, high operational costs to serve many borrowers for small loans and high risk exposure to companies with limited credit history. Addressing these challenges, financial technology company Nikel (<https://nikel.com/>) makes it easier for banks to lend to SMEs through automation and analytics on its digital credit platform, powered by Google Cloud (<https://cloud.google.com/>).

“Digital lending is inevitable, but banks often don’t have the infrastructure to act quickly,” says Reinier Musters, founder and Chief Executive Officer at Nikel. “Banks can leverage our digital credit platform hosted on Google Cloud to support SMEs, without spending time to build the IT system or the partner ecosystem.”

“To solve the SME credit gap with technology, we may need to make a sideways or lateral move in terms of our product,” says Austin. “Vault

Core on Google
Cloud gives us
the flexibility and
flavor to offer
new services by
adding modules
to our core
platform without
buying
additional
software
licenses.”

—Jonathan Austin, Chief
Technology Officer, Nikel

Accelerating SME lending innovation on a cloud-native core banking platform

Indonesia is the primary focus for Nikel, where more than 64 million SMEs

(<https://setkab.go.id/en/12-million-msmes-available-in-digital-ecosystem-minister-says/>)

contribute to 61% of GDP and employ 91% of the workforce.

“Thanks to Google Cloud’s presence in Indonesia, we can meet data sovereignty rules and quickly set up our operations to help SMEs,” says Jonathan (Jon) Austin, Chief Technology Officer at Nikel. “One

notable example is helping pharmacies get loans faster to meet the unprecedented demand for medical supplies during the COVID-19 pandemic.”

Nikel implemented a micro-services modern technology stack that consists of Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>) (GKE) and Vault Core by Thought Machine

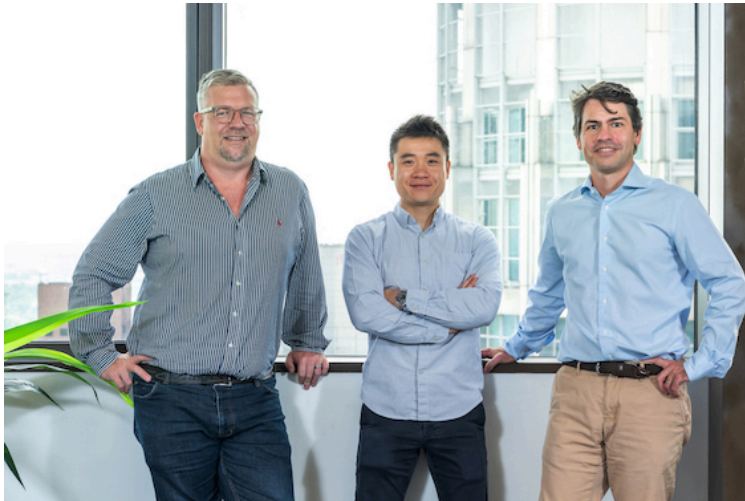
(<https://thoughtmachine.net/>) — a cloud-native modern core banking system. The company also uses CloudSQL to store transaction data and machine learning (ML) algorithms by BigQuery ([/bigquery](https://cloud.google.com/bigquery/)) for credit risk analysis. Powered by Vault Core, Nikel is able to gain access to transaction data in real time and house it on Cloud SQL

(<https://cloud.google.com/sql/>), making it possible to use BigQuery's robust ML algorithms to automate credit risk analysis and create a frictionless customer experience for its SME customers.

“To solve the SME credit gap with technology, we may need to make a sideways or lateral move in terms of our product,” says Austin. “Vault Core on Google Cloud gives us the flexibility and speed to offer new services by adding unique product offerings to our core platform without buying additional software licenses or having to rely on the core provider.”

Thought Machine was instrumental in helping Nikel deploy and configure Vault Core in three months. Since then, Nikel has defined and deployed additional innovative products as market opportunity arises. Vault Core’s microservices-based

platform and Universal Product Engine allow Nikel to launch new products and services quickly while reaping benefits of cloud computing in scalability, agility, and cost.



Jonathon Austin (CTO), Mackenzie Tan (co-founder, COO & Head of Legal), Reinier Musters (founder & CEO)

“When a
borrower applies
for a loan
through a fintech
or digital
platform that we
support, we can
predict the
default risk with

BigQuery and
deliver a loan
recommendation
in milliseconds.”

—Jonathan Austin, Chief
Technology Officer, Nikel

Driving down the cost of SME lending with streamlined workflows

Typically banks need to check multiple documents from borrowers such as identification documents and company records to comply with Know Your Customer (KYC) regulations in the financial industry. The large amount of paperwork required for SME lending has been hurdle for loan officers and slowing down the approval process to the point of negatively impacting customer satisfaction. Nikel removes the lending friction by automating document processing at scale.

It was vital for Nikel to digitize the KYC process, particularly during the COVID-19 pandemic when loan officers couldn't conduct in-person verification. Keeping their customer's health and safety in mind, Nikel incorporated e-signatures and e-contracts to facilitate digital loan documents.

Nikel uses a microservices architecture to speed up development time by reusing a common code for

deploying existing capabilities to different partners.

“Within six months of adopting Google Kubernetes Engine, we launched five products including e-signatures and cut the time it takes them to develop and launch a new financial product by half each time,” says Austin.

Furthermore, being able to automatically integrate certified electronic signature providers in Indonesia with the Nikel system on GKE allows banks to identify more than 30 million potential customers. This now enables loan officers to focus on understanding customer needs and recommending the right products instead of chasing down missing information and reentering data from KYC documents in the correct format.

“Currently, we query hundreds of millions of records each day, from model development to monitoring. BigQuery can easily scale to billions of rows of data as we

expand our
business and
provide our
lending partners
with predictive
analysis to
better manage
SME credit
risks.”

—Jonathan Austin, Chief
Technology Officer, Nikel

Improving risk decisions with machine learning and BigQuery

Credit decision-making is often considered by banks the most challenging stage of a loan life cycle. To help ease the process, Nikel analyzed and anonymized the transaction dataset of four million SME customers with the help of BigQuery to understand loan repayment patterns and built ML models to predict the likelihood of default.

“It is common for SMEs to wait two to three weeks for banks to decide on loan applications,” says Austin. “However, when a borrower applies for a loan through a fintech or digital platform that we support, we can predict the default risk with BigQuery and deliver a loan recommendation in milliseconds.”

Nikel also combines other data sources such as bank transaction data for accurate credit decisions, which is more predictive of a borrower's credit risk than deposits alone. If the borrower doesn't meet the bank's acceptable level of default risk, the credit model recommends a reduced loan amount to the originator instead of an outright rejection.

From a data science perspective, Nikel continually monitors the performance of the ML models and updates them as new scenarios present themselves.

"Currently, we query hundreds of millions of records each day, from model development to monitoring," Austin adds. "BigQuery can easily scale to billions of rows of data as we expand our business and provide our lending partners with predictive analysis to better manage SME credit risks."

Musters says that as technology plays a pivotal role in changing the future of SME lending, he is optimistic about the company's vision of closing the SME credit gap by helping banks improve lending processes and credit analytics on our platform. "We certainly want to do more with our Google Cloud products such as GKE and BigQuery, to accelerate innovation and help more SMEs get the funds they need to grow and contribute to the economy."

